8 Assignment 3

9 Introduction

In this report for the thrid assignment we will take on the tasks of analyzing the two Laser Interferometer Gravitational-Wave-Observatory (LIGO) instruments.

10 Task 1

I read both of the files containing the data from the LIGO detectors Hanford and Livingford by using the `h5py` module. I found that both the Hanford signal $x_H[n]$ and the Livingstone signal $x_L[n]$ had 131072 samples. I used the sample rate of $f_s = 4096$ and calculated that each signal was 32 seconds long (See Appendix A)

11 Task 2

I implemented the library `matplotlib.pyplot` to plot both of the signals into a subplot that is shown below.

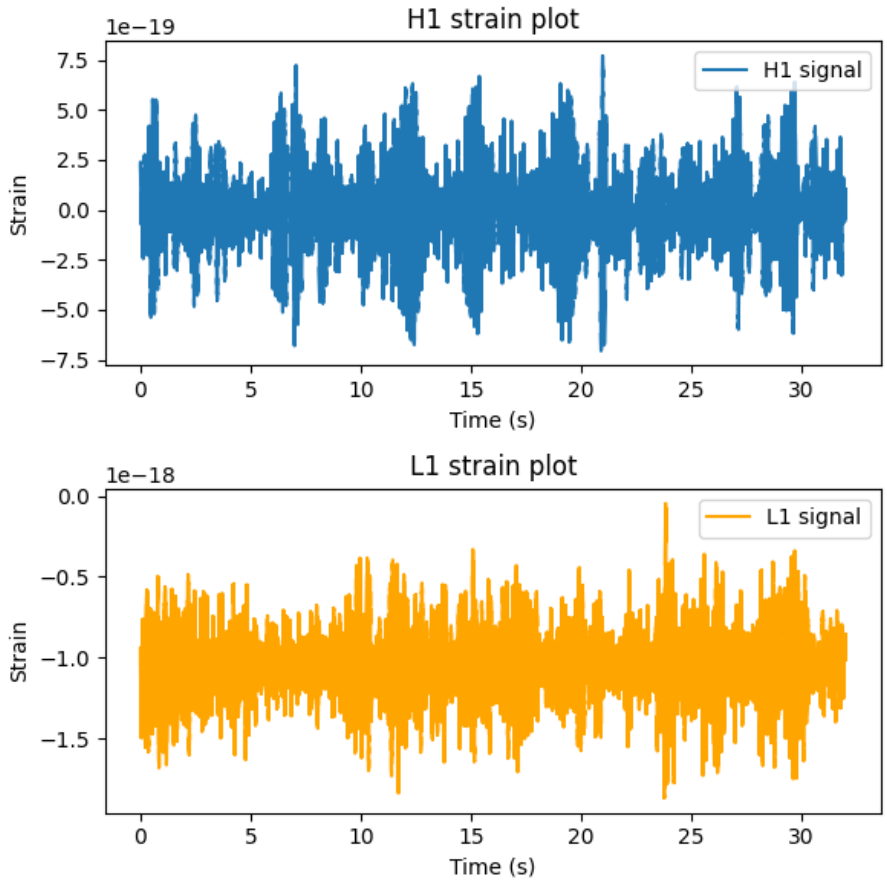


Figure 7: Plot of the LIGO Hanford and Livingstone signal

From the x-axis it is clear that the signal lasts for 32 seconds. The y-axis is represents the strain of the signal which varies in a range of $-7.04 * 10^{-19}$ to $7.7 * 10^{-19}$ for the Hanford signal and for the Livingford signal it spans from $-1.87 * 10^{-19}$ to $-4.6 * 10^{-20}$. This is the stretching of space caused by the passage of gravitational wave through the instrument.

12 Task 3

12.1 a)

I imported the tapered functions `hann` and `hamming` from the `scipy.signal` module and used the `hann` window for the following tasks.

12.2 b)

From the assignment we have the test signal:

$$x[n] = 10^{-5} cos(\frac{2\pi f_1 n}{f_s}) + cos(\frac{2\pi f_1 n}{f_s})$$

consisting of two sinusioid with a strong and a weak amplitude and the values.

```
#Sample rate
f_s = 4096
#Frequency 1
f_1 = 31.5
#Frequency 2
f_2 = 1234.56
#Sample signal length
N = 4096
```

The plot for the original and hann windowed test signal.

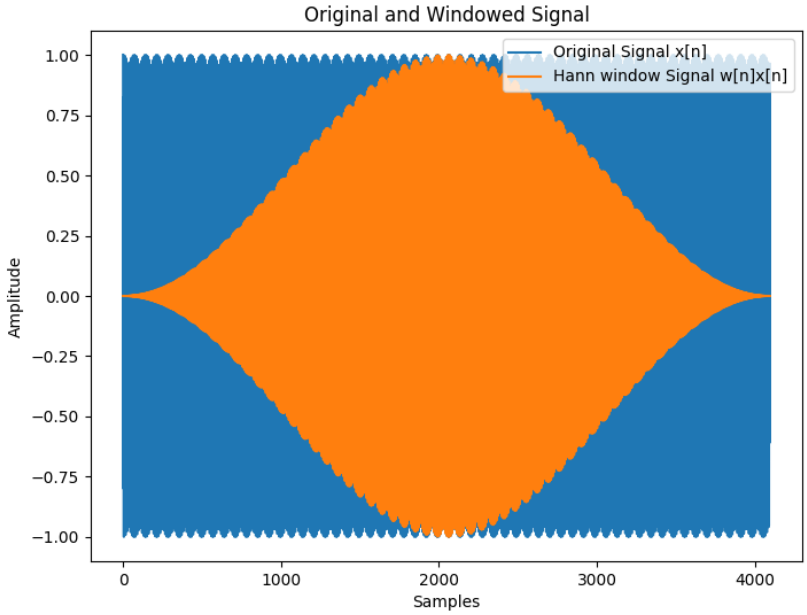


Figure 8: Original and windowed signal for test signal

From this plot we can see how the tapered Hann window modifies the signal. The original test signal appears squared and extends uniformly across the sample range, with sharp edges that creates discontinuity and spectral leakage. The Hann window reduces this by minimizing the abrupt edges, it has its maximum

amplitude in the middle and smoothly goes towards zero at the start and end of the signal $n = 0$ and $n = 4095$. It focuses the central position and the true frequencies in the middle.

12.3 c)

From the principle spectrum $-f_s/2 < f_k < f_s/2$ we can check with a sample frequency of $f_s = 4096$ that the frequency spectrum spans from -2048 to 2048 Hz.

$$-\frac{4096}{2} < f_k < \frac{4096}{2}$$
$$-2048 < f_k < 2048$$

The FFT corresponds to angular frequency of $\omega_k = \frac{2\pi k}{N}$ for each index k. We can convert this frequency to hertz in the principle spectrum by writing

$$\begin{aligned} f_k &= \frac{\omega_k}{2\pi} * f_s \\ &= \frac{\frac{2\pi k}{N}}{2\pi} f_s \\ &= \frac{k}{N} f_s \end{aligned}$$

12.4 d)

I computed the FFT of the both the signals with the numpy function `numpy.fft.fft` and then calculated the power of each frequency component and converted to decible by the equation $10log_{10}(|\hat{x}[k]|^2)$. I then plotted the power spectrum for the signal for positive and negative frequencies.

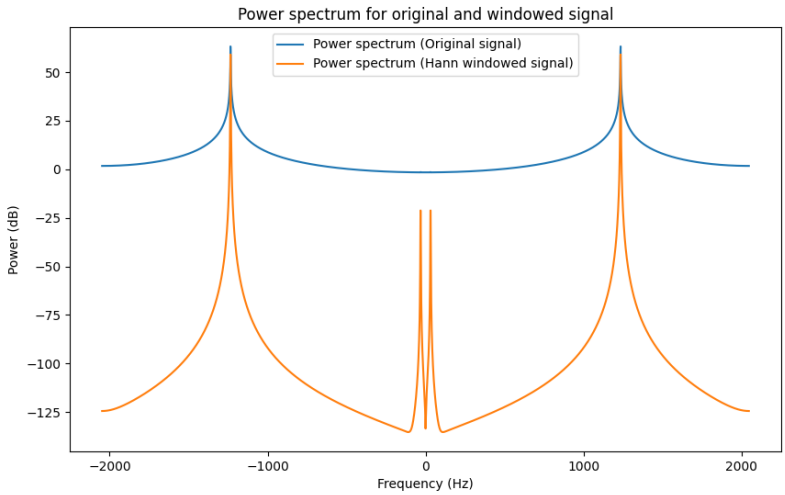


Figure 9: Power spectrum of windowed test signal (tapered and untapered)

12.5 e)

I have highlighted the locations on both the original and windowed signal where it intercepts the specified frequencies of $\pm 32.5Hz$ and $\pm 1234.56Hz$. and included the markers in the plot beneath.

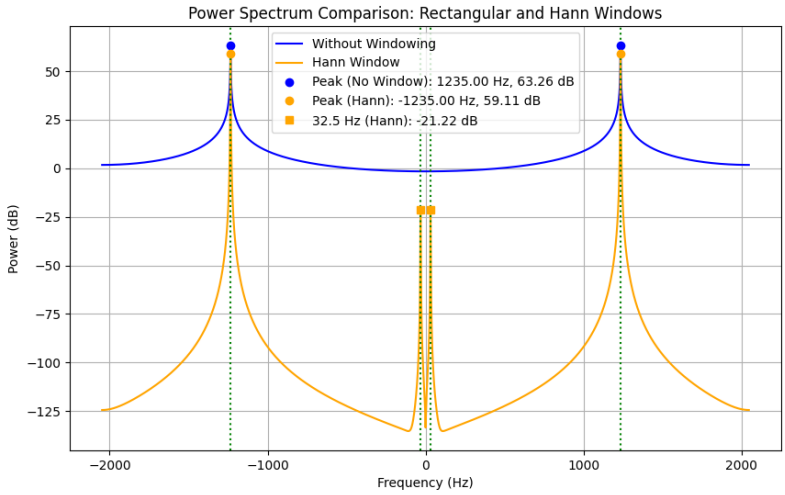


Figure 10: Power spectral response with highlighted frequencies

12.6 f)

For the Hann window signal theres a peak for on the 31.5Hz frequency, but the original signal is almost flat at this x-axis and therefore does not allow us to detect the 31.5 Hz spectral lines

12.7 g)

I made a plot for the hamming window and plotted it comparing to the hann window.

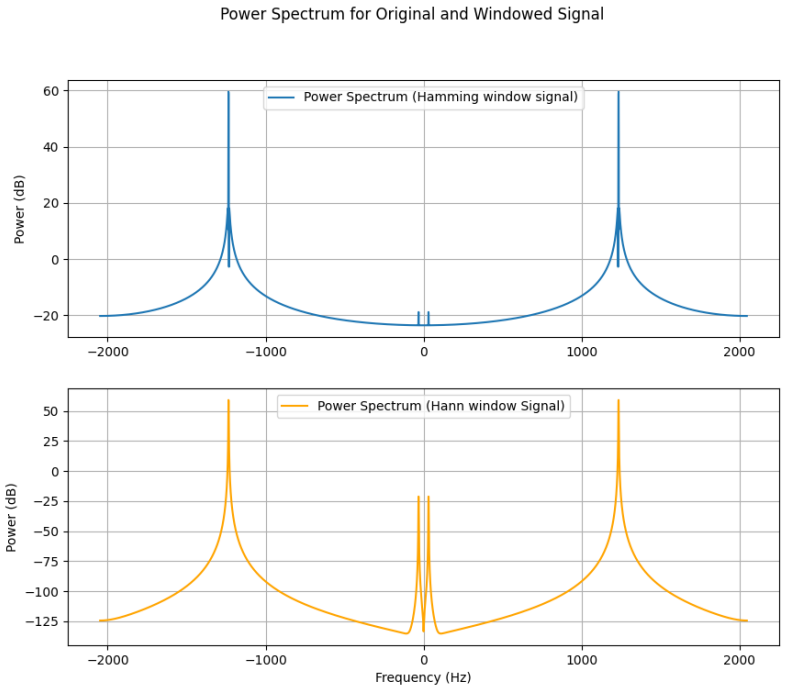


Figure 11: Hamming and Hann windowed signal plot

The Hann window is best at detecting the weaker signals and therefore i will continue using this for the next section.

13 Task 4

I calculated the power spectrum of the LIGO signals as seen in Appendix C. From this we can calculate the power spectrum for every frequency. I plotted the result with power (dB) on the vertical axis and the frequency (Hz) on the horizontal axis.

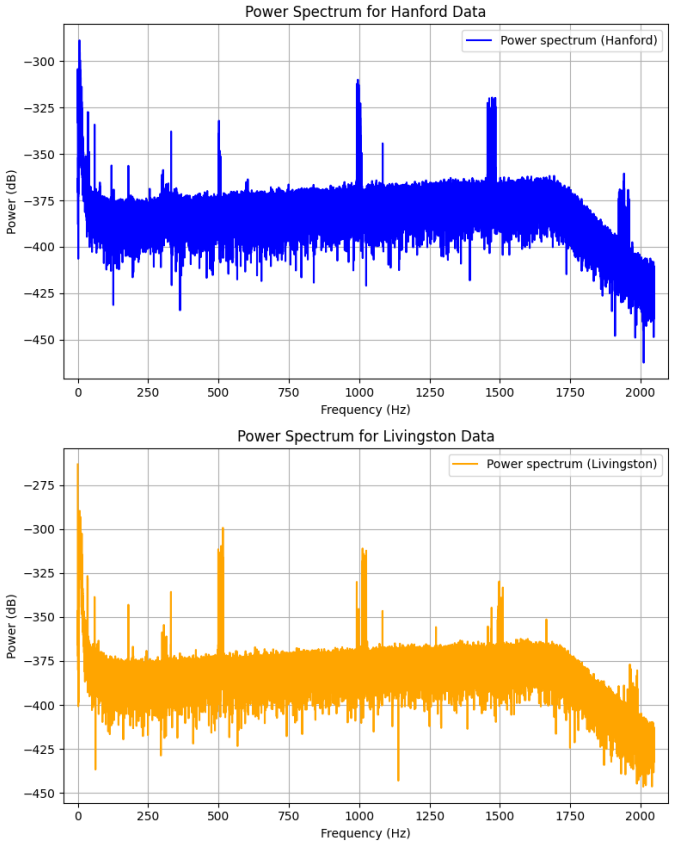


Figure 12: Power spectrum of windowed LIGO signals

Power spectrum with highlighted strong spectrum components:

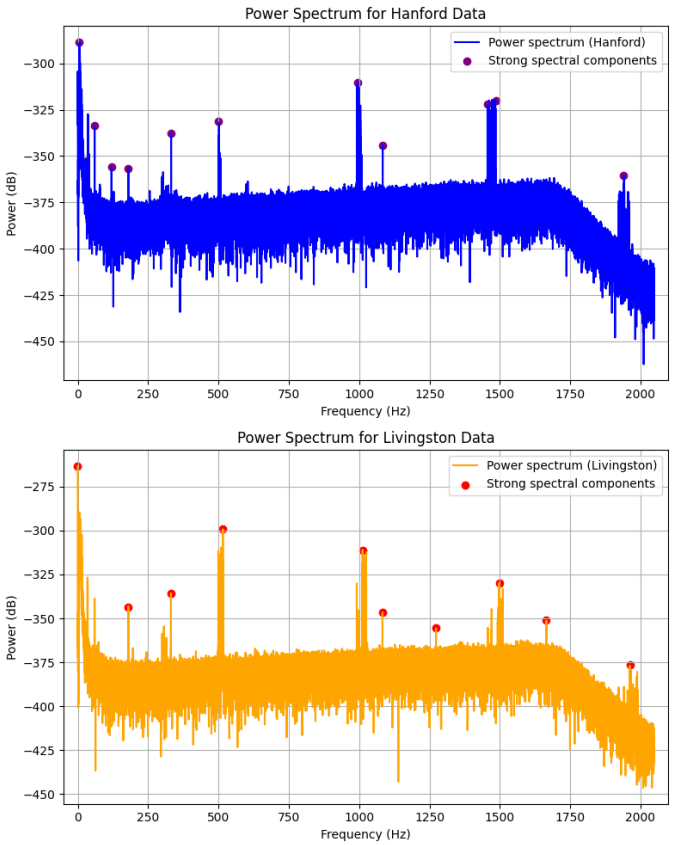


Figure 13: Power spectrum with marked strong spectral components

To plot these points i zoomed into the different chosen peaks and listed the positions in x and y-coordinates in form of frequency (Hz) and power (dB). These frequencies and power of the strong spectral components are listed here:

```
strong_freq_h = [5.4, 59.7, 121.7, 178.6, 332.2, 500.1, 996.1,
    1084.4,1456.2, 1485.5, 1940.1]
strong_power_h = [-288.6, -333.6, -355.8, -356.7, -337.7, -331.3,
    -310.2, -344.1, -322.0, -320.0, -360.6]

strong_freq_L= [-1.4, 179.4, 330.5, 515.6, 1012.2, 1083.5, 1273.7,
    1497.8, 1665.0, 1963.0]
strong_power_L = [-263.1, -343.4, -335.9, -299.2, -311.2, -346.8,
    -355.4, -329.8, -351.1, -376.4]
```

14 Task 5

I implemented two separate whitening filters in the frequency domain $\hat{h}[k] = \frac{1}{\hat{x}[k]}$ and used an inverse discrete fourier transform in python with function `numpy.fft.ifft` to transform the whitened signal into a time-domain. Here is the plot i obtained for the whitened singal for the Hanford and Livingford signal.

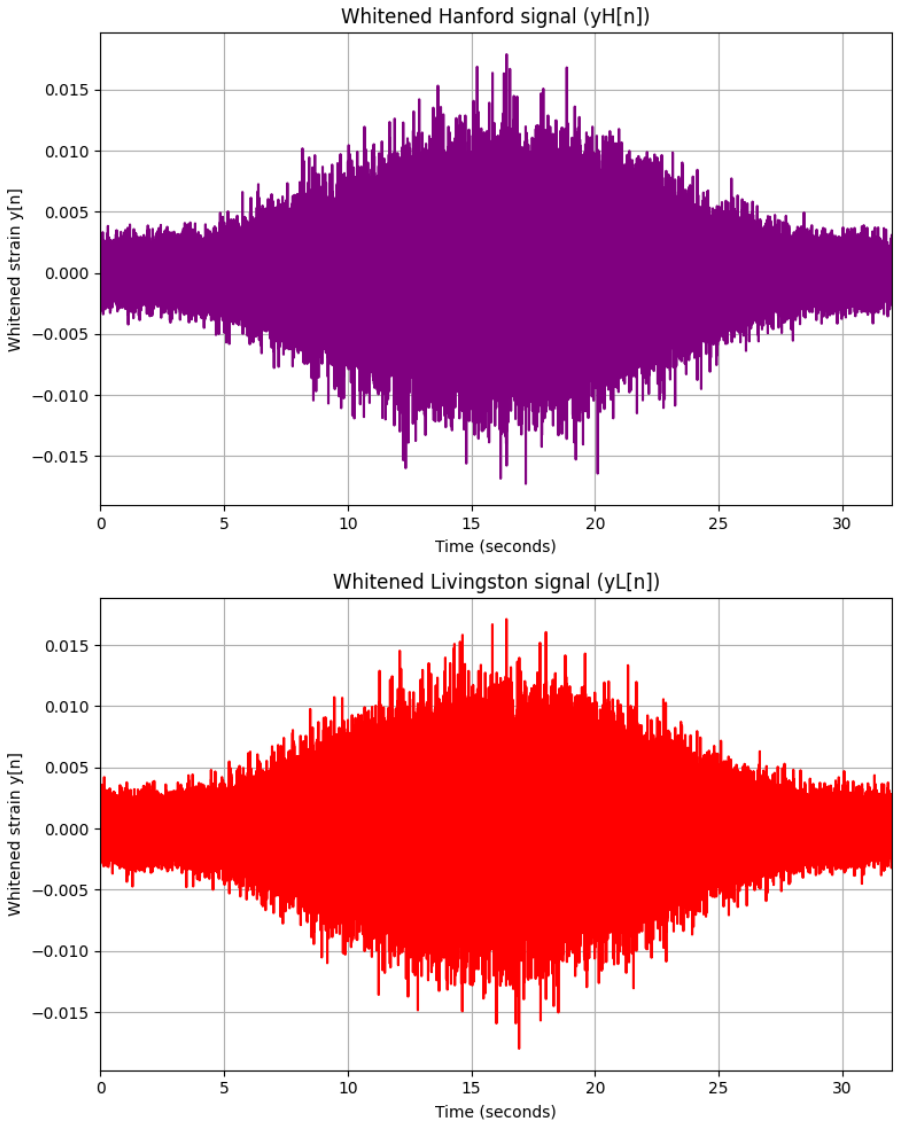


Figure 14: Whitened signal for Handford and Livingstone signal

I plotted the signals for the total time of 32 seconds and we can see that the gravitational wave signal happens in the middle between 16.2 and 16.5 seconds

15 Task 6

15.1 a)

I tried finding an integer L so the filter will reduce the power of frequency at $f = 300Hz$ with -6dB by using the formula $L = \frac{f_s}{2*f_{cutoff}}$ where f_s is the sample rate and f_{cutoff} is 300Hz. Substituting the values it gave a result of $L = 7$ which didn't look exactly correct on the plot. I attempted the values close to 7 as an integer and found the best result using $L = 8$ as shown in the plot beneath.

15.2 b)

I plotted the power spectral response of the filter in dB and got the result:

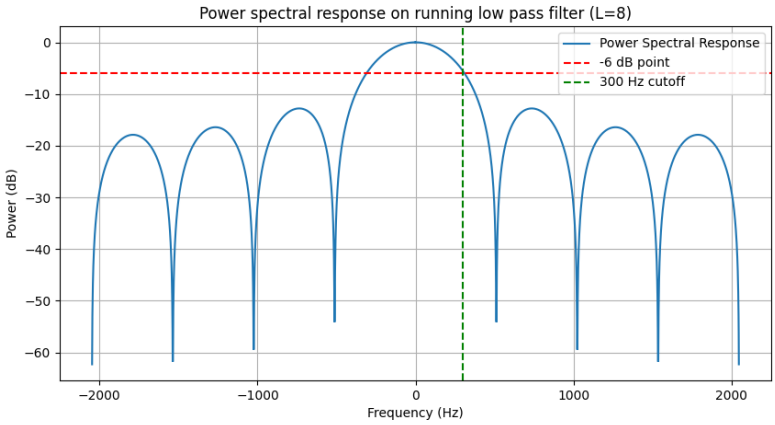


Figure 15: power spectral response of low pass filter in dB scale

15.3 c)

I calculated the time delay using the following equation

$$\tau = \frac{(L_{filter} - 1)}{2f_s}$$

I found the result for τ with the integer L to be $8.54 * 10^{-4}$.

15.4 d, e, f)

I applied the running mean average low pass filter to the whitened LIGO signals Hanford and Livington using function `numpy.convolve` and shifted signal in time by finding the time length of the signal and adding the time delay. I plotted the whitened signals with the simple low pass filter for the time interval [16.1, 16.6] where we found the gravitational wave signal

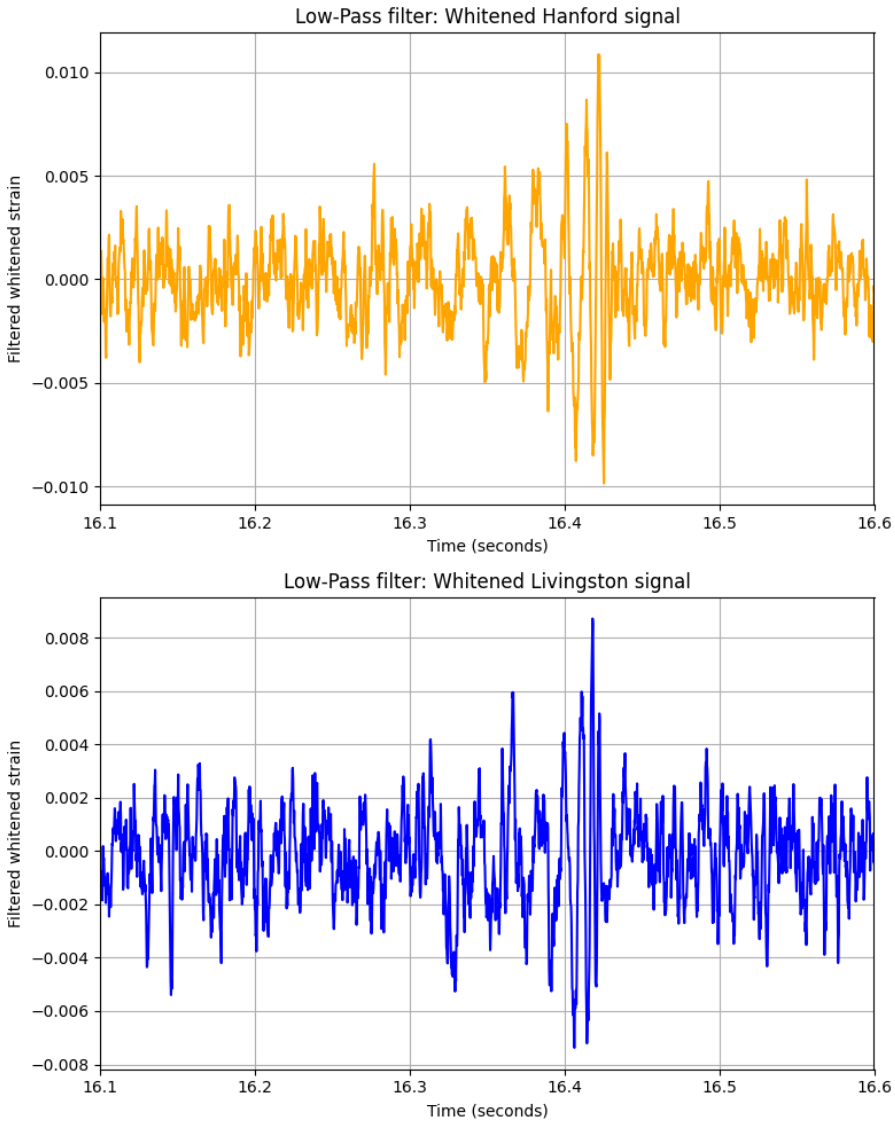


Figure 16: Whitened low pass filter for Hanford and Livingstone signal

Additionally i plotted them in the same graph.

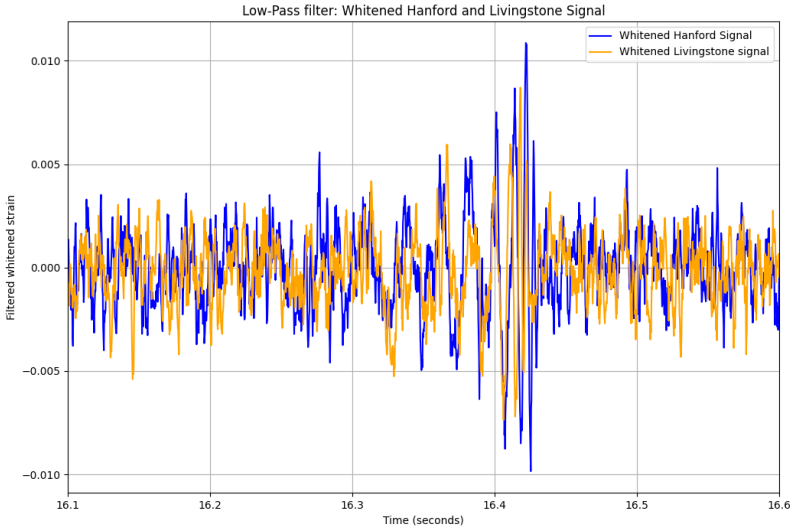


Figure 17: Whitened low pass filter for Hanford and Livingstone signal in same plot

16 Task 7

16.1 a, b)

To determine time seperation between the two signals i wanted to plot the magnitude of the filtered gravitational wave signals where one of the time signals had delay n_0 which corresponded to a time delay in a range of [-10ms, 10ms]. I tried finding this unknown deleay n_0 where the signals were approximately aligned by using a function `correlation` that computes cross correlation between the two signals to find the delay that gives the maximum similarity, but didnt get a good result. I started trying different values for n_0 until i found that the two signals were best aligned for $n_0 = 19$. I obtained the plot under.

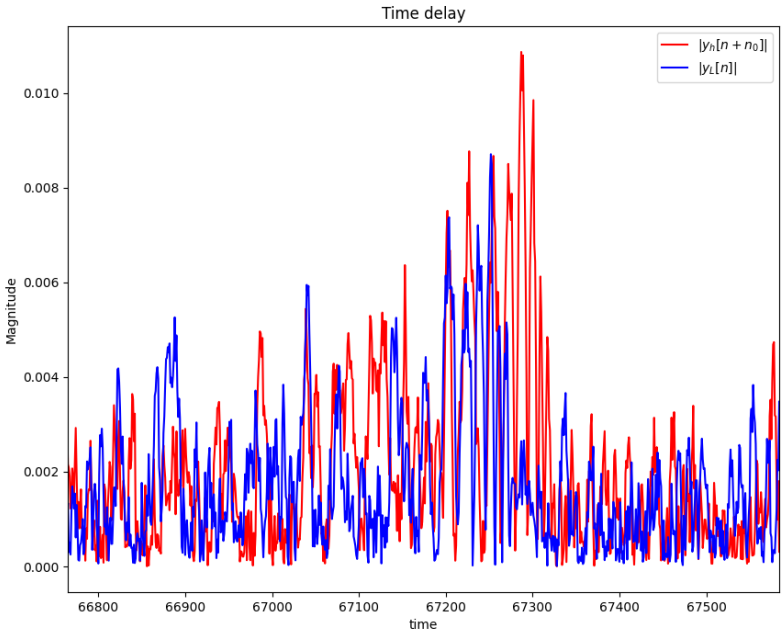


Figure 18: Plotted magnitude of filtered signals $y_h[n]$ and $y_L[n]$

I identified the optimal alignments of the two plots when the peaks of the time-shifted magnitude of the Hanford signal fit closely with the Livingford signal.

16.2 c)

This value for n_0 correlates to a time-delay value of $-4.64 * 10^{-3}$ which was calculated by taking $\tau = \frac{n_0}{f_s}$.

16.3 d)

This value for the time delay τ is well within the limit and therefore agrees with the gravitational-wave propagation speed.

17 Task 8

I calculated the dynamic spectrum of the low-pass filtered whitening signal.

I plotted the dynamic power spectrum in dB scale for both of the LIGO signals between the time range of 15.5 and 17 seconds.

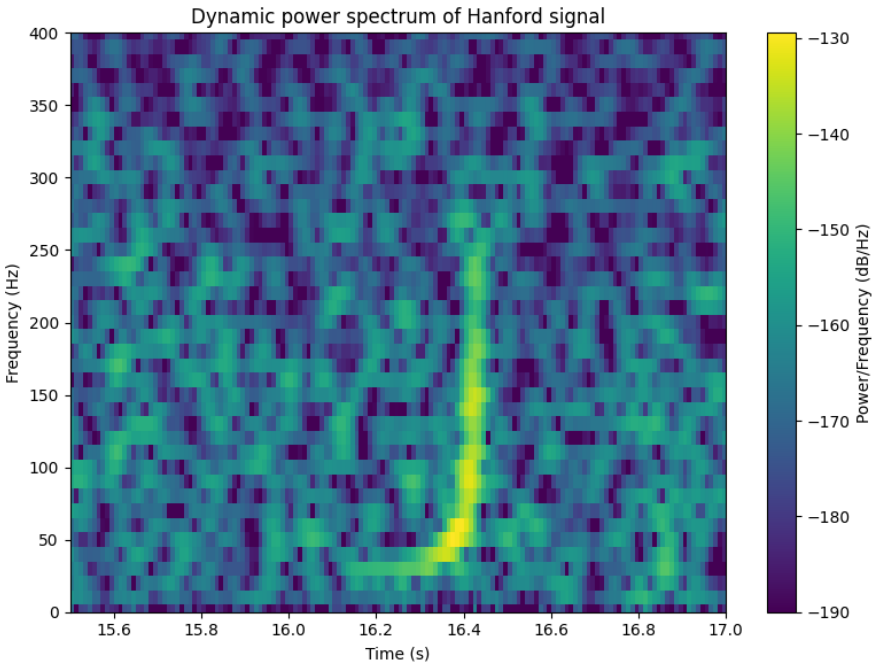


Figure 19: Dynamic power spectrum fro Hanford signal

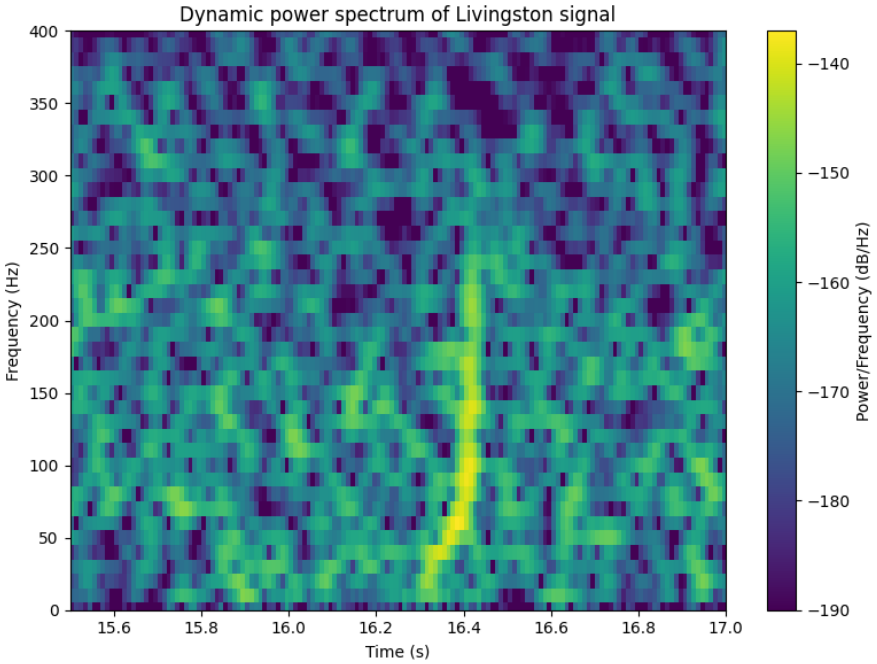


Figure 20: Dynamic power spectrum for Livingford signal

References

[1] Jørn Olav Vierinen, Juha; Jensen. *Signal Processing.* UiT, Tromsø, Norge, 2024.